Unique Ergodicity for the stochastic skew heat equation

Oleg Butkovsky Jonathan C. Mattingly Lorenzo Zambotti

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Overview of Proof of Main Theorem

We will prove the uniqueness of the invariant measure by showing that the Markov Semigroup associated with the system is Asymptotic Strong Feller. We will do this by constructing asymptotic coupling of the two equations beginning from arbitrary, different initial conditions u_0 and v_0. We do this by introducing an intermediate process $\widetilde{v}_t$ whose law on path space is equivalent to v_t but which converges to u_t as $t \to \infty$.

This is done by in introducing a drift term of the form $\lambda_t(u_t - \widetilde{v}_t)$ to the equation for $\widetilde{v}_t$ so the equation for $\|u_t - \widetilde{v}_t\|^2$ has a contracting term. We essentially pick λ_t to be of comparable size to $\|u_t - \widetilde{v}_t\|^{-\beta}$ for some $\beta > 0$. This makes the equation strongly contractive. Since we will pick λ_t to be a piecewise constant, $\|u_t - \widetilde{v}_t\| \to 0$ only asymptotically as $t \to \infty$. To establish the needed absolue continuity on the infinite time horizon via Girsanov's theorem, we will have to show that the appropriate infinite-time square integrality holds.

After the contractiveness and the integrability, the main technical lemma of the proof is Lemma 9. This lemma requires some stochastic sewing estimates.

Proof of Main Theorem

Let $D := [0,1]$, p is the Dirichlet heat kenrel. For $x \in D$, $u_0, v_0 \in C_0([0,1])$, $t > 0$, we define $V(t,x) := \int_0^t \int_D p_{t-r}(x,y) W(dr,dy)$,

$$u_t(x) = P_t u_0(x) + \int_0^t \int_D p_{t-r}(x,y) h(u_r(y))\, dydr + V_t(x), \tag{1}$$

$$v_t(x) = P_t v_0(x) + \int_0^t \int_D p_{t-r}(x,y) h(v_r(y))\, dydr + V_t(x), \tag{2}$$

$$\widetilde{v}_t(x) = P_t v_0(x) + \int_0^t \int_D p_{t-r}(x,y) \big(h(\widetilde{v}_r(y)) + \lambda_r(u_r(y) - \widetilde{v}_r(y))\big)\, dydr + V_t(x). \tag{3}$$

Here $\lambda\colon [0,\infty) \to \mathbb{R}_+$ is a deterministic function which is piecewise constant on intervals $[n, n+1)$ for $n \in \mathbb{N}$; $h\colon \mathbb{R} \to \mathbb{R}$ is a smooth function with $\|h\|_{C^{-1}} \leqslant 1$.

If $f\colon \Omega \times [S,T] \times D \to \mathbb{R}$ is a measurable function, then for $m \geqslant 1$ we define

$$\|f\|_{\mathcal{C}^{0,0}L_m([S,T])} := \sup_{s\in[S,T]} \sup_{x \in D} \|f_s(x)\|_{L_m(\Omega)}. \tag{4}$$

We state the following technical result which we prove later in the text.

Lemma 1. *Let $m \geqslant 2$, $\mu \in (0, \frac{1}{6})$, Then there exists a constant $C > 0$ which depends only on m, such that for any $n \in \mathbb{N}$, $n \leqslant s \leqslant t \leqslant n+1$, $x, z \in D$ we have*

$$\Big\| \int_s^t \int_D e^{-\lambda_n(t-r)} p_{t-r}(x,y)(h(u_r(y)) - h(\widetilde{v}_r(y)))\, dydr \Big\|_{L_m(\Omega)} \leqslant C \lambda_n^{-\frac{1}{8}} \big(\lambda_n^{-1} + \|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([s,t])}\big). \tag{5}$$

$$\Big\| \int_s^t \int_D e^{-\lambda_n(t-r)} (p_{t-r}(x,y) - p_{t-r}(z,y))(h(u_r(y)) - h(\widetilde{v}_r(y)))\, dydr \Big\|_{L_m(\Omega)}$$
$$\leqslant C|x-z|^{\mu} \big(\lambda_n^{-\frac{9}{8}} + \|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([s,t])}\big). \tag{6}$$

Now we proceed to the main part of the proof.

Lemma 2. *Let $n \in \mathbb{N}$, $t \in [n, n+1)$, $x \in D$. Then we have*

$$u_t(x) - \widetilde{v}_t(x) = e^{-\lambda_n(t-n)} P_{t-n}(u_n - \widetilde{v}_n)(x)$$
$$+ \int_n^t \int_D e^{-\lambda_n(t-r)} p_{t-r}(x,y) \big(h(u_r(y)) - h(\widetilde{v}_r(y))\big)\, dydr. \tag{7}$$

Proof. Let $t \in [n, n+1)$, $x \in D$ and $T > t$. Then

$$P_{T-t}(u_t - \widetilde{v}_t)(x) = P_{T-n}(u_n - \widetilde{v}_n)(x) + \int_n^t \int_D p_{T-r}(x,y) \big(h(u_r(y)) - h(\widetilde{v}_r(y)) - \lambda_n(u_r(y) - \widetilde{v}_r(y))\big)\, dydr.$$

Therefore, using the chain rule (in t for fixed $x \in D$) we derive

$$e^{\lambda_n t} P_{T-t}(u_t - \widetilde{v}_t)(x) = e^{n\lambda_n} P_{T-n}(u_n - \widetilde{v}_n)(x) + \lambda_n \int_n^t e^{\lambda_n r} P_{T-r}(u_r - \widetilde{v}_r)(x)\, dr$$
$$+ \int_n^t \int_D e^{\lambda_n r} p_{T-r}(x,y) \big(h(u_r(y)) - h(\widetilde{v}_r(y)) - \lambda_n(u_r(y) - \widetilde{v}_r(y))\big)\, dydr$$
$$= e^{n\lambda_n} P_{T-n}(u_n - \widetilde{v}_n)(x) + \int_n^t \int_D e^{\lambda r} p_{T-r}(x,y) \big(h(u_r(y)) - h(\widetilde{v}_r(y))\big)\, dydr.$$

Thus, by letting $T \searrow t$ and using continuity of u, $\widetilde{v}$ we get (7). $\square$

Lemma 3. *Let $m \geqslant 2$. There exists a universal constant $C = C(m)$ with the following property: if $\inf_{n\in\mathbb{N}} \lambda_n \geqslant C$, then for any $n \in \mathbb{N}$ we have*

$$\|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([n,n+1])} \leqslant 2 \sup_{x\in D} \|u_n(x) - v_n(x)\|_{L_m(\Omega)} + C\lambda_n^{-\frac{9}{8}}.$$

Proof. For any $t > 0$, $x \in D$ and any measurable function $f\colon \Omega \times D \to \mathbb{R}$,

$$\|P_t f(x)\|_{L_m(\Omega)} \leqslant \int_D p_t(x,y)\|f(y)\|_{L_m(\Omega)}\, dy \leqslant \sup_{y\in D} \|f(y)\|_{L_m(\Omega)}. \tag{8}$$

Using this inequality, (7) and taking $s = n$ in (5), we derive for $t \in [n, n+1]$, $x \in D$

$$\|u_t(x) - \widetilde{v}_t(x)\|_{L_m(\Omega)} \leqslant \sup_{x\in D} \|u_n(x) - v_n(x)\|_{L_m(\Omega)} + C\lambda_n^{-\frac{9}{8}} + C\lambda_n^{-\frac{1}{8}}\|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([n,n+1])}$$

for $C = C(m) > 0$. Therefore, taking in the above inequality supremum over $t \in [n, n+1]$, $x \in D$, we get

$$\|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([n,n+1])} \leqslant \sup_{x\in D} \|u_n(x) - v_n(x)\|_{L_m(\Omega)} + C\lambda_n^{-\frac{9}{8}} + C\lambda_n^{-\frac{1}{8}}\|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([n,n+1])}.$$

We suppose that for all $n \in \mathbb{N}$ we have $C\lambda_n^{-\frac{1}{8}} \leqslant \frac{1}{2}$. Then we get (8). □

Next, we improve bound (8).

Lemma 4. *Let $m \geqslant 2$. There exist universal constants $\Gamma_0 = \Gamma_0(m)$, $\Gamma = \Gamma(m) \geqslant \Gamma_0^2$ such that if for any $n \in \mathbb{N}$ we have $\lambda_n > \Gamma_0$ then for any $n \in \mathbb{N}$ we have*

$$\sup_{x\in D} \|u_{n+1}(x) - \widetilde{v}_{n+1}(x)\|_{L_m(\Omega)} \leqslant \frac{1}{4} \sup_{x\in D} \|u_n(x) - \widetilde{v}_n(x)\|_{L_m(\Omega)} + \Gamma\lambda_n^{-\frac{9}{8}}.$$

Proof. Fix $n \in \mathbb{N}$. It follows from (7), inequality (8) and Lemma 1, that for any $x \in D$

$$\begin{aligned}
\|u_{n+1}(x) - \widetilde{v}_{n+1}(x)\|_{L^m(\Omega)} &\leqslant e^{-\lambda_n} \sup_{x\in D} \|u_n(x) - \widetilde{v}_n(x)\|_{L^m(\Omega)} + C\lambda_n^{-\frac{1}{8}}(\lambda^{-1} + \|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L^m([n,n+1])}) \\
&\leqslant e^{-\lambda_n} \sup_{x\in D} \|u_n(x) - \widetilde{v}_n(x)\|_{L^m(\Omega)} + C\lambda_n^{-\frac{9}{8}} + \\
&\quad + 2C\lambda_n^{-\frac{1}{8}}\big(\sup_{x\in D} \|u_n(x) - v_n(x)\|_{L^m(\Omega)} + C\lambda_n^{-\frac{9}{8}}\big).
\end{aligned}$$

Then we need that $e^{-\lambda_n} + 2C\lambda_n^{-\frac{1}{8}} < \frac{1}{4}$ to conclude. □

Finally, we pass on to the analysis on $[0, \infty)$. Fix $m \geqslant 2$. Given u_0, v_0, we define λ_n, $n \in \mathbb{Z}_+$, as the smallest real number such that $\Gamma\lambda_n^{-\frac{9}{8}} \leqslant 2^{-n-2}\sup_{x\in D}|u_0(x) - v_0(x)|$ and $\lambda_n \geqslant \Gamma_0$ where $\Gamma = \Gamma(m)$, $\Gamma_0 = \Gamma_0(m)$ are from Lemma 4.

Lemma 5. *There exists $C > 0$ such that for any $n \in \mathbb{N}$ we have*

$$\sup_{x\in D} \|u_n(x) - \widetilde{v}_n(x)\|_{L_m(\Omega)} \leqslant 2^{-n}\|u_0 - v_0\|_\infty; \tag{9}$$

$$d_{TV}(\mathrm{Law}(v_n), \mathrm{Law}(\widetilde{v}_n)) \leqslant C\|u_0 - v_0\|_\infty^{\frac{1}{9}}; \tag{10}$$

$$\|\sup_{x\in D} |u_n(x) - \widetilde{v}_n(x)|\,\|_{L_m(\Omega)} \leqslant C2^{-n}\|u_0 - v_0\|_\infty; \tag{11}$$

Proof. First, let us show (9). We proceed by induction. Indeed, if $n = 0$, (9) is obviously satisfied. Assume (9) holds for $n \in \mathbb{N}$. Then, by Lemma 4,

$$\sup_{x\in D} \|u_{n+1}(x) - \widetilde{v}_{n+1}(x)\|_{L_m(\Omega)} \leqslant \frac{1}{4} \sup_{x\in D} \|u_n(x) - \widetilde{v}_n(x)\|_{L_m(\Omega)} + \Gamma\lambda_n^{-\frac{9}{8}} \leqslant 2^{-n-1}\|u_0 - v_0\|_\infty$$

and thus (9) holds. To show (10), we note that if $\|u_0 - v_0\|_\infty \geqslant 1$ then (10) is obvious. Otherwise, if $\|u_0 - v_0\|_\infty \leqslant 1$, recalling that $\Gamma \geqslant \Gamma_0^2$ by Lemma 4, we see that $\Gamma\lambda_n^{-\frac{9}{8}} = 2^{-n-2} \sup_{x\in D} |u_0(x) - v_0(x)|$. Note that thanks to Lemma 3 an (9), we have for any $t \geqslant 0$

$$\sup_{x\in D} \|u_t(x) - \widetilde{v}_t(x)\|_{L_m(\Omega)} \leqslant 2^{-t+2} \sup_{x\in D} |u_0(x) - v_0(x)|. \tag{12}$$

Therefore

$$\begin{aligned}\int_0^\infty \int_D \lambda_t^2 \mathsf{E}|u_t(x) - \widetilde{v}_t(x)|^2 \,dxdt &\leqslant \int_0^\infty \lambda_t^2 \sup_{x\in D} \mathsf{E}|u_t(x) - \widetilde{v}_t(x)|^2 \,dt \\ &\leqslant \Gamma^2 \sup_{x\in D} |u_0(x) - v_0(x)|^{\frac{2}{9}} \int_0^\infty 2^{-\frac{2}{9}t+8} \,dt = C \sup_{x\in D} |u_0(x) - v_0(x)|^{\frac{2}{9}}.\end{aligned}$$

where Γ is from Lemma 4. Bound (10) follows now from Pinsker's inequality and the Girsanov's theorem, see, e.g., [BKS20, Lemma A.1, Theorem A.2]. Finally, to show (11), we use Lemma 1 and Lemma 2. Let $\mu \in (0, \frac{1}{6})$. We derive for any $x, z \in D$

$$\begin{aligned}&\|u_{n+1}(x) - \widetilde{v}_{n+1}(x) - (u_{n+1}(z) - \widetilde{v}_{n+1}(z))\|_{L_m(\Omega)} \leqslant \int_D |p_1(x,y) - p_1(z,y)|\|u_n(y) - \widetilde{v}_n(y)\|_{L_m(\Omega)} \,dy \\ &\quad + \Big\|\int_n^{n+1} \int_D e^{-\lambda_n(n+1-r)}(p_{n+1-r}(x,y) - p_{n+1-r}(z,y))(h(u_r(y)) - h(\widetilde{v}_r(y))) \,dydr\Big\|_{L_m(\Omega)} \\ &\leqslant C|x-z|^\mu\big(\|u - \widetilde{v}\|_{\mathcal{C}^{0,0}L_m([n,n+1])} + \lambda_n^{-\frac{9}{8}}\big)\end{aligned}$$

for $C = C(m, \mu)$. Taking now $\mu = \frac{1}{7}$ and recalling (12) and the definition of λ_n, we get for any $x, z \in D$

$$\|u_{n+1}(x) - \widetilde{v}_{n+1}(x) - (u_{n+1}(z) - \widetilde{v}_{n+1}(z))\|_{L_m(\Omega)} \leqslant C|x-z|^{\frac{1}{7}} 2^{-n}\|u_0 - v_0\|_\infty$$

for $C = C(m)$. Choosing now m large enough (it is sufficient to take $m > 7$), and applying the Kolmogorov continuity theorem, we obtain (11). □

Proof of the main result. Let $\mathcal{P}^h$ is the semigroup associated with the solution to (1). Consider a state space $C_0([0,1])$ equipped with the distance $|x - y|_\infty$. Then for any function $\varphi : C_0([0,1]) \to \mathbb{R}$, $u_0, v_0 \in C_0([0,1])$, $n \in \mathbb{N}$ we have by Lemma 5

$$\begin{aligned}|\mathcal{P}_n^h\varphi(u_0) - \mathcal{P}_n^h\varphi(v_0)| &= |\mathsf{E}\varphi(u_n) - \mathsf{E}\varphi(v_n)| \leqslant |\mathsf{E}\varphi(\widetilde{v}_n) - \mathsf{E}\varphi(v_n)| + |\mathsf{E}\varphi(u_n) - \mathsf{E}\varphi(\widetilde{v}_n)| \\ &\leqslant \|\varphi\|_{L_\infty(C_0([0,1])} d_{TV}(\mathrm{Law}(v_n), \mathrm{Law}(\widetilde{v}_n)) + \|\varphi\|_{Lip(C_0([0,1]))} \mathsf{E}\|u_n - \widetilde{v}_n\|_\infty^{\frac{1}{9}} \\ &\leqslant C\|\varphi\|_{L_\infty(C_0([0,1])}\|u_0 - \widetilde{v}_0\|_\infty^{\frac{1}{9}} + C2^{-\frac{n}{9}}\|u_0 - \widetilde{v}_0\|_\infty^{\frac{1}{9}}.\end{aligned}$$

Taking now $h = g_{1/m}$ and passing to the limit as $m \to \infty$, we see that semigroup associated to the skew stochastic heat equation is asymptotically strong Feller (see [HM06] for the corresponding definitions). Arguing exactly as in [BW20, proof of Theorem 2.7], we see that (10) and (11) imply that $\mathcal{P}$ is irreducible. Therefore by [HM06, Corollary 3.17] (see also [BW20, Theorem 2.8]), $\mathcal{P}$ has no more than one invariant measure. □

It remains now to prove Lemma 1. We will use stochastic sewing.

For $0 \leqslant S < T$, consider the simplices $\Delta_{[S,T]} := \{(s,t) \in [S,T]^2 \colon s < t\}$ and $\Delta^3_{[S,T]} := \{(s,u,t) \in [S,T]^3 \colon s < u < t\}$. If $A_{\cdot,\cdot}$ is a function $\Delta_{[S,T]} \to \mathbb{R}^d$, then we put $\delta A_{s,u,t} := A_{s,t} - A_{s,u} - A_{u,t}, \quad (s,u,t) \in \Delta^3_{[S,T]}$. Let $(\mathcal{F}_t)_{t\in[0,T]}$ be the filtration generated by the Wiener process defining V and denote by $\mathsf{E}^t[\cdot] := \mathsf{E}[\cdot|\mathcal{F}_t]$ the conditional expectation given $\mathcal{F}_t$, $t \in [0,T]$. For $x \in D$, $(s,t) \in \Delta_{[0,1]}$ denote $\sigma^2(s,t,x) = \mathsf{E}(V(t,x) - \mathsf{E}^s V(t,x))^2$.

Lemma 6. *There exists $C > 0$ such that for any $s,t \in \Delta_{[0,1]}$, $x \in D$ we have $\sigma^2(s,t,x) \geqslant C((1-x) \wedge x \wedge (t-s)^{1/2})$.*

Proof. Fix $x \in D$. Suppose without loss of generality that $x \leqslant 1-x$. We have $\mathsf{E}(V(t,x) - \mathsf{E}^s V(t,x))^2 \geqslant \frac{1}{2}\int_0^{t-s} p_r(x,x)\,dr$.For $r > 0$, using the Fourier representation of p_r on has $p_r(x,x) \geqslant \frac{e^{-2\pi^2}}{4\sqrt{r}}$ if $x^2 \geqslant r$ and $r \leqslant \frac{1}{16}$. Therefore $\mathsf{E}(V(t,x) - \mathsf{E}^s V(t,x))^2 \geqslant \frac{1}{2}\int_0^{(t-s)\wedge x^2 \wedge \frac{1}{16}} p_r(x,x)\,dr \geqslant C((t-s)^{1/2} \wedge x)$, which implies the statement of the lemma. □

Lemma 7. *For $\rho \in (-2,0]$, $\mu \in (0,1]$, there exists $C = C(\rho,\mu) > 0$ so that for any $a \in [0,1]$, $\kappa \in (0,1]$, $x,z \in D$ we have*

$$\int_D p_\kappa(x,y)(y \wedge (1-y) \wedge a)^\rho\,dy \leqslant C a^\rho (1 + \frac{a^2}{\kappa});$$
$$\int_D |p_\kappa(x,y) - p_\kappa(z,y)|(y \wedge (1-y) \wedge a)^\rho\,dy \leqslant C|x-z|^\mu \kappa^{-\frac{\mu}{2}} a^\rho (1 + \frac{a^2}{\kappa}).$$

Proof. The bounds follows by a direct calculation using the spectral formula for the Dirichlet heat kernel and the fact that for any $\mu > -1$ there exists a constant $C = C(\mu) > 0$ such that for any $\kappa \in [0,1]$ we have $\sum_{n=1}^\infty e^{-n^2\kappa} n^\mu \leqslant C\kappa^{-\frac{1}{2}-\frac{\mu}{2}}$. □

The next corollary follows from Lemma 6 and that Lemma 7

Corollary 8. *Let $\rho \in (-4,0]$, $\mu \in (0,1]$. Then there exists $C = C(\rho,\mu) > 0$ such that for any $x,z \in D$, $(s,t) \in \Delta_{[0,1]}$, $\kappa \in [t-s,1]$, we have $\int_D p_\kappa(x,y)\sigma(s,t,y)^\rho\,dy \leqslant C(t-s)^{\frac{\rho}{4}}$ and $\int_D |p_\kappa(x,y) - p_\kappa(z,y)|\sigma(s,t,y)^\rho\,dy \leqslant C|x-z|^\mu (t-s)^{\frac{\rho}{4}-\frac{\mu}{2}}$.*

Now we can derive some key integral bounds, which eventually imply Lemma 1. For a function $\psi \colon \Omega \times [S,T] \times D \to \mathbb{R}$, $\tau \in (0,1]$, $m \geqslant 1$ we consider the following Hölder-type seminorm

$$[f]_{\mathcal{C}^{\tau,0}L_m[S,T]} := \sup_{s,t\in\Delta_{[S,T]}} \sup_{x\in D} \frac{\|f(t,x) - \mathsf{E}^s f(t,x)\|_{L_m(\Omega)}}{(t-s)^\tau};.$$

For brevity, we denote $\mathcal{X} := \mathcal{C}^{\frac{3}{4},0}L_m([S,T])$.

Lemma 9. *Let $m \geqslant 2$. Let $w \colon [0,1] \to \mathbb{R}$ be a continuous function, $\mu \in (0,\frac{1}{6})$. Then there exists a constant $C = C(\mu,m) > 0$ such that for any bounded continuous function $b \colon \mathbb{R} \to \mathbb{R}$, any continuous adapted processes $\varphi, \psi \colon \Omega \times [0,1] \times D \to \mathbb{R}$, $x,z \in D$, and any $0 \leqslant S \leqslant T \leqslant T_0 \leqslant 1$ with $T - S \leqslant T_0 - T$ setting $\Gamma := C\|b\|_{\mathcal{C}^{-1}}\|w\|_{L_\infty([S,T])}$,*

$$\mathcal{A}^\varphi_t(x) := \int_0^t \int_D w_r\, p_{T_0-r}(x,y)\, b(V_r(y) + \varphi_r(y))drdy, \quad t \in [0,T], \tag{13}$$

and setting $\mathcal{A}^{\varphi,\psi}_t(x) := \mathcal{A}^{\varphi}_t(x) - \mathcal{A}^{\psi}_t(x)$, $\mathcal{A}^{\varphi,\psi}_{s,t}(x) := \mathcal{A}^{\varphi,\psi}_t(x) - \mathcal{A}^{\varphi,\psi}_s(x)$ *we have*

$$\|\mathcal{A}^{\varphi}_T(x) - \mathcal{A}^{\varphi}_S(x)\| \leqslant \Gamma(T-S)^{\frac{3}{4}}\big(1 + [\varphi]_{\mathcal{X}}(T-S)^{\frac{1}{2}}\big); \tag{14}$$

$$\|\mathcal{A}^{\varphi,\psi}_{S,T}(x)\|_{L_m(\Omega)} \leqslant \Gamma(T-S)^{\frac{7}{12}}\|\psi-\varphi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0}L_m([S,T])} + \Gamma(T-S)^{\frac{5}{4}}\big([\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big); \tag{15}$$

$$\begin{aligned}\|\mathcal{A}^{\varphi,\psi}_{S,T}(x) - \mathcal{A}^{\varphi,\psi}_{S,T}(z)\|_{L_m(\Omega)} \leqslant{}& \Gamma|x-z|^{\mu}(T-S)^{\frac{7}{12}-\frac{\mu}{2}}\|\psi-\varphi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0}L_m([S,T])}\\ &+ \Gamma|x-z|^{\mu}(T-S)^{\frac{5}{4}-\frac{\mu}{2}}\big([\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big).\end{aligned} \tag{16}$$

Proof. Fix $(S,T,T_0) \in \Delta^3_{[0,1]}$, $x \in D$. We begin with the proof of (14). We use the stochastic sewing lemma (SSL) [Lê20, Theorem 2.1] and consider the germ

$$A^{\varphi}_{s,t} := \mathsf{E}^s \int_s^t \int_D w_r p_{T_0-r}(x,y) b(V(r,y) + \mathsf{E}^s\varphi(r,y))\, drdy, \quad (s,t) \in \Delta_{[S,T]}. \tag{17}$$

We claim that for any $(s,u,t) \in \Delta_{[S,T]}$ we have

$$\|A^{\varphi}_{s,t}\|_{L_m(\Omega)} \leqslant \Gamma(t-s)^{\frac{3}{4}}; \quad \|\mathsf{E}^s\delta A^{\varphi}_{s,u,t}\|_{L_m(\Omega)} \leqslant \Gamma[\varphi]_{\mathcal{X}}(t-s)^{\frac{5}{4}}. \tag{18}$$

We see that (18) implies that all the assumptions of [Lê20, Theorem 2.1] are satisfied. Thus after checking (18), the desired bound (14) immediately follows from [Lê20, Theorem 2.1].

We begin with the first part of (18). We have for any $(s,t) \in \Delta_{[S,T]}$

$$\begin{aligned}|A^{\varphi}_{s,t}| &\leqslant \int_s^t \int_D w_r p_{T_0-r}(x,y) |\mathsf{E}^s b(V(r,y) + \mathsf{E}^s\varphi(r,y))|\, drdy\\ &= \int_s^t \int_D w_r p_{T_0-r}(x,y) |G_{\sigma^2(s,r,y)} b(\mathsf{E}^s V(r,y) + \mathsf{E}^s\varphi(r,y))|\, drdy\\ &\leqslant \|w\|_{L_\infty([S,T])} \int_s^t \int_D p_{T_0-r}(x,y) \|G_{\sigma^2(s,r,y)} b\|_{L_\infty(\mathbb{R})}\, drdy\\ &\leqslant C\|w\|_{L_\infty([S,T])}\|b\|_{\mathcal{C}^{-1}} \int_s^t \int_D p_{T_0-r}(x,y)(\sigma(s,r,y))^{-1}\, drdy \leqslant \Gamma(t-s)^{\frac{3}{4}},\end{aligned}$$

where in the last line we used Corollary 8 with $\kappa = T_0 - r$, $\rho = -1$. Note that $\kappa \geqslant T_0 - T \geqslant T - S \geqslant t - s$ and thus the conditions of Corollary 8 hold. This shows the first part of (18).

To check the second part of (18), we write for $(s,u,t) \in \Delta^3_{[S,T]}$

$$\begin{aligned}|\mathsf{E}^s\delta A^{\varphi}_{s,u,t}| &\leqslant \mathsf{E}^s \int_u^t \int_D w_r p_{T_0-r}(x,y) |\mathsf{E}^u b(V(r,y) + \mathsf{E}^s\varphi(r,y)) - \mathsf{E}^u b(V(r,y) + \mathsf{E}^u\varphi(r,y))|\, drdy\\ &\leqslant \mathsf{E}^s \int_u^t \int_D w_r p_{T_0-r}(x,y) \|G_{\sigma^2(u,r,y)} b\|_{\mathcal{C}^1} |\mathsf{E}^s\varphi(r,y)) - \mathsf{E}^u\varphi(r,y))|\, drdy\\ &\leqslant \Gamma\mathsf{E}^s \int_u^t \int_D p_{T_0-r}(x,y)(\sigma(u,r,y))^{-2} |\mathsf{E}^s\varphi(r,y)) - \mathsf{E}^u\varphi(r,y))|\, drdy.\end{aligned}$$

Therefore,

$$\|\mathsf{E}^s\delta A^{\varphi}_{s,u,t}\|_{L_m(\Omega)} \leqslant \Gamma[\varphi]_{\mathcal{X}}(t-s)^{\frac{3}{4}} \int_u^t \int_D p_{T_0-r}(x,y)(\sigma(u,r,y))^{-2}\, drdy \leqslant \Gamma[\varphi]_{\mathcal{X}}(t-u)^{\frac{5}{4}},$$

where in the last inequality we used Corollary 8 with $\kappa = T_0 - r$, $\rho = -2 > -4$. We see again that $\kappa \geqslant T_0 - T \geqslant T - S \geqslant t - s$ and thus the conditions of Corollary 8 hold. This implies the second part of (18). Thus, all the conditions of the stochastic sewing lemma are satisfied and (14) follows from [Lê20, Theorem 2.1].

Equation (15) follows again from the SSL. Consider the germ $A_{s,t} := A^{\varphi}_{s,t} - A^{\psi}_{s,t}$ and process $\mathcal{A}_t := \mathcal{A}^{\varphi}_t - \mathcal{A}^{\psi}_s$, for $(s,t) \in \Delta_{[S,T]}$ where A^{φ}, $\mathcal{A}^{\varphi}$ were define previously and A^{ψ}, $\mathcal{A}^{\psi}$ are defined analogously. We claim now that for any $(s,u,t) \in \Delta_{[S,T]}$ we have

$$\|A_{s,t}\|_{L_m(\Omega)} \leqslant \Gamma \|\varphi - \psi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0}L_m([S,T])} (t-s)^{\frac{7}{12}}; \; \|\mathsf{E}^s \delta A_{s,u,t}\|_{L_m(\Omega)} \leqslant \Gamma([\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}})(t-s)^{\frac{5}{4}}. \quad (19)$$

We see again that (19) implies that all the conditions of the SSL are satisfied. We also see that the second part of (19) immediately follows from the second part of (18). Thus, it remains only to verify the first part of (19). We argue similarly to the first part of the proof and get for any $(s,t) \in \Delta_{[S,T]}$

$$\begin{aligned}
|A_{s,t}| &\leqslant \int_s^t \int_D w_r p_{T_0-r}(x,y) |\mathsf{E}^s b(V(r,y) + \mathsf{E}^s \varphi(r,y)) - \mathsf{E}^s b(V(r,y) + \mathsf{E}^s \psi(r,y))| \, dr dy \\
&= \int_s^t \int_D w_r p_{T_0-r}(x,y) |G_{\sigma^2(s,r,y)} b(\mathsf{E}^s V(r,y) + \mathsf{E}^s \varphi(r,y)) - G_{\sigma^2(s,r,y)} b(\mathsf{E}^s V(r,y) + \mathsf{E}^s \psi(r,y))| \, dr dy \\
&\leqslant \|w\|_{L_\infty([S,T])} \int_s^t \int_D p_{T_0-r}(x,y) \|G_{\sigma^2(s,r,y)} b\|_{\mathcal{C}^{\frac{2}{3}}(\mathbb{R})} |\mathsf{E}^s \varphi(r,y) - \mathsf{E}^s \psi(r,y)|^{\frac{2}{3}} \, dr dy \\
&\leqslant \Gamma \int_s^t \int_D p_{T_0-r}(x,y) (\sigma(s,r,y))^{-\frac{5}{3}} |\mathsf{E}^s \varphi(r,y) - \mathsf{E}^s \psi(r,y)|^{\frac{2}{3}} \, dr dy.
\end{aligned}$$

Hence, using again Corollary 8 with $\kappa = T_0 - r$, $\rho = -2$, we derive

$$\begin{aligned}
\|A_{s,t}\|_{L_m(\Omega)} &\leqslant \Gamma \int_s^t \int_D p_{T_0-r}(x,y) (\sigma(s,r,y))^{-\frac{5}{3}} \|\mathsf{E}^s \varphi(r,y) - \mathsf{E}^s \psi(r,y)\|^{\frac{2}{3}}_{L_m(\Omega)} \, dr dy \\
&\leqslant \Gamma \|\varphi - \psi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0}L_m([S,T])} \int_s^t \int_D p_{T_0-r}(x,y) (\sigma(s,r,y))^{-\frac{5}{3}} \, dr dy \\
&\leqslant \Gamma \|\varphi - \psi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0}L_m([S,T])} (t-s)^{\frac{7}{12}}.
\end{aligned}$$

This shows the first part of (19). Thus, all the conditions of the stochastic sewing lemma are satisfied and (15) follows from (19). Finally, we show (22). We use again the stochastic sewing lemma, which we apply similarly to the germ

$$A_{s,t} := \mathsf{E}^s \int_s^t \int_D (p_{T_0-r}(x,y) - p_{T_0-r}(z,y)) w_r (b(V(r,y) + \mathsf{E}^s \varphi(r,y)) - b(V(r,y) + \mathsf{E}^s \psi(r,y))) \, dr dy,$$

where $(s,t) \in \Delta_{[S,T]}$ and the process $\mathcal{A}^{\varphi,\psi}_t(x)$. Arguing exactly as in the first part of the proof and using Corollary 8, we derive for any $(s,u,t) \in \Delta_{[S,T]}$ $\|A_{s,t}\|_{L_m(\Omega)} \leqslant \Gamma |x - z|^{\mu} \|\varphi - \psi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0}L_m([S,T])} (t-s)^{\frac{7}{12} - \frac{\mu}{2}}$ and $\|\mathsf{E}^s \delta A^{\varphi}_{s,u,t}\|_{L_m(\Omega)} \leqslant \Gamma([\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}) |x - z|^{\mu} (t-s)^{\frac{5}{4} - \frac{\mu}{2}}$Since $\frac{7}{12} - \frac{\mu}{2} > \frac{1}{2}$ and $\frac{5}{4} - \frac{\mu}{2} > 1$, the conditions of SSL are satisfied, implying (22). □

Corollary 10. *Suppose that all the conditions of Lemma 9 are satisfied. Then there exists a constant $C = C(m) > 0$ such that bounds (14), (15), (22) hold for any $0 \leqslant S \leqslant T \leqslant 1$, $T_0 = T$ (without extra restriction $T - S \leqslant T_0 - T$).*

Proof. The result follows immediately from Lemma 9 and the taming singularities lemma [BFG21, Lemma 2.3]. ∎

Corollary 11. *Let $m \geqslant 2$. Let $w\colon [0,1] \to \mathbb{R}$ be a continuous function, $\mu \in (0, \frac{1}{6})$. Then there exists a constant $C = C(\mu, m) > 0$ such that for any bounded continuous function $b\colon \mathbb{R} \to \mathbb{R}$, any continuous adapted processes $\varphi, \psi\colon \Omega \times [0,1] \times D \to \mathbb{R}$, $x, z \in D$, and any $0 \leqslant S \leqslant T \leqslant 1$ we have*

$$\begin{aligned}
&\Big\| \int_S^T \int_D w_r p_{T-r}(x,y) \Big(b(V(r,y) + \varphi(r,y)) - b(V(r,y) + \psi(r,y)) \Big)\, dr dy \Big\|_{L_m(\Omega)} \\
&\quad \leqslant C \|b\|_{\mathcal{C}^{-1}} \|w\|_{L_\infty([S,T])} (T-S)^{\frac{1}{4}} \|\psi - \varphi\|_{\mathcal{C}^{0,0} L_m([S,T])} \\
&\qquad + C \|b\|_{\mathcal{C}^{-1}} \|w\|_{L_\infty([S,T])} (T-S)^{\frac{5}{4}} \big(1 + [\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big); \qquad (20)
\end{aligned}$$

$$\begin{aligned}
&\Big\| \int_S^T \int_D w_r (p_{T-r}(x,y) - p_{T-r}(z,y)) (b(V(r,y) + \varphi(r,y)) - b(V(r,y) + \psi(r,y)))\, dr dy \Big\|_{L_m(\Omega)} \\
&\quad \leqslant C \|b\|_{\mathcal{C}^{-1}} \|w\|_{L_\infty([S,T])} |x-z|^{\mu} \|\psi - \varphi\|_{\mathcal{C}^{0,0} L_m([S,T])} \\
&\qquad + C \|b\|_{\mathcal{C}^{-1}} \|w\|_{L_\infty([S,T])} |x-z|^{\mu} (T-S)^{\frac{5}{4} - \frac{\mu}{2}} \big(1 + [\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big)
\end{aligned}$$

Proof. The first inequality follows from Corollary 10, bound (15) and the Young inequality $a_1 a_2 \leqslant a_1^{\frac{3}{2}} + a_2^3$ applied to $a_1 := (T-S)^{\frac{1}{6}} \|\psi - \varphi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0} L_m([S,T])}$, $a_2 := (T-S)^{\frac{5}{12}}$. The second inequality follows in the same way, this time applying the Young inequality to $a_1 := \|\psi - \varphi\|^{\frac{2}{3}}_{\mathcal{C}^{0,0} L_m([S,T])}$, $a_2 := (T-S)^{\frac{7}{12} - \frac{\mu}{2}}$. ∎

Corollary 12. *Let $m \geqslant 2$, $\mu \in (0, \frac{1}{6})$. Then there exists a constant $C = C(\mu, m) > 0$ such that for any bounded continuous function $b\colon \mathbb{R} \to \mathbb{R}$, any continuous adapted processes $\varphi, \psi\colon \Omega \times [0,1] \times D \to \mathbb{R}$, $x, z \in D$, $\lambda > 1$, and any $(s,t) \in \Delta_{[0,1]}$ we have*

$$I^{\lambda}_{s,t}(x) := \int_s^t \int_D e^{-\lambda(t-r)} p_{t-r}(x,y) \left(b(V_r(y) + \varphi_r(y)) - b(V_r(y) + \psi_r(y)) \right) dr dy$$

$$\|I^{\lambda}_{s,t}(x)\|_{L_m(\Omega)} \leqslant C \|b\|_{\mathcal{C}^{-1}} \lambda^{-\frac{1}{8}} \|\varphi - \psi\|_{\mathcal{C}^{0,0} L_m([s,t])} + C \|b\|_{\mathcal{C}^{-1}} \lambda^{-\frac{9}{8}} \big(1 + [\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big); \qquad (21)$$

$$\begin{aligned}
\|I^{\lambda}_{s,t}(x) - I^{\lambda}_{s,t}(z)\|_{L_m(\Omega)} &\leqslant C \|b\|_{\mathcal{C}^{-1}} |x-z|^{\mu} \|\psi - \varphi\|_{\mathcal{C}^{0,0} L_m([s,t])} \\
&\quad + C \|b\|_{\mathcal{C}^{-1}} |x-z|^{\mu} \lambda^{-\frac{9}{8}} \big(1 + [\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big). \qquad (22)
\end{aligned}$$

Proof. Fix $\lambda > 1$, let $\varepsilon > 0$ be a parameter, which we will choose later. We split the integral in the left hand side of (21) into two: over the time interval $[s, (t - \lambda^{-1+\varepsilon}) \vee s]$ and over the interval $[(t - \lambda^{-1+\varepsilon}) \vee s, t]$. If $r \leqslant t - \lambda^{-1+\varepsilon}$, then $e^{-\lambda(t-r)} \leqslant e^{-\lambda^{\varepsilon}}$ and $\|e^{-\lambda(t-\cdot)}\|_{L_\infty([s,(t-\lambda^{-1+\varepsilon}) \vee s])} \leqslant e^{-\lambda^{\varepsilon}}$. Therefore, Corollary 11 implies

$$\begin{aligned}
&\Big\| \int_s^{(t-\lambda^{-1+\varepsilon}) \vee s} \int_D e^{-\lambda(t-r)} p_{t-r}(x,y) \Big(b(V(r,y) + \varphi(r,y)) - b(V(r,y) + \psi(r,y)) \Big)\, dr dy \Big\|_{L_m(\Omega)} \\
&\quad \leqslant C \|b\|_{\mathcal{C}^{-1}} e^{-\lambda^{\varepsilon}} \big(1 + \|\varphi - \psi\|_{\mathcal{C}^{0,0} L_m([s,t])} + [\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big). \qquad (23)
\end{aligned}$$

for $C = C(m)$. On the other hand, if $r \in [(t - \lambda^{-1+\varepsilon}) \vee s, t]$, then $e^{-\lambda(t-r)} \leqslant 1$ and Corollary 11 implies

$$\begin{aligned}
&\Big\| \int_{(t-\lambda^{-1+\varepsilon}) \vee s}^t \int_D e^{-\lambda(t-r)} p_{t-r}(x,y) \Big(b((V + \varphi)(r,y)) - b((V + \psi)(r,y)) \Big)\, dr dy \Big\|_{L_m(\Omega)} \\
&\quad \leqslant C \|b\|_{\mathcal{C}^{-1}} \lambda^{-\frac{1}{4}(1-\varepsilon)} \|\varphi - \psi\|_{\mathcal{C}^{0,0} L_m([s,t])} + C \|b\|_{\mathcal{C}^{-1}} \lambda^{-\frac{5}{4}(1-\varepsilon)} \big(1 + [\varphi]_{\mathcal{X}} + [\psi]_{\mathcal{X}}\big). \qquad (24)
\end{aligned}$$

for $C = C(m)$. Choose now ε such that $\frac{5}{4}(1-\varepsilon) = \frac{9}{8}$. Then (21) follows from (23), (24), taking into account that for such ε and some universal $C > 0$ we have $\exp(-\lambda^\varepsilon) \leqslant C\lambda^{-\frac{9}{8}}$ for all $\lambda \geqslant 1$. Bound (22) follows from Corollary 11 by exactly the same argument. $\square$

Lemma 13. *Let $m \geqslant 2$. Then there exists a constant $C = C(m) > 0$ such that for any bounded continuous function $b\colon \mathbb{R} \to \mathbb{R}$, a continuous in space and $\mathcal{F}_0$-measurable process $z\colon \Omega \times D \to \mathbb{R}$, any continuous adapted processes $Z, \rho\colon \Omega \times [0,1] \times D \to \mathbb{R}$ satisfying*

$$Z(t,x) = P_t z(x) + \int_0^t \int_D p_{t-r}(x,y)\big(b(Z(r,y)) + \rho(r,y)\big)\, dydr + V(t,x), t \in [0,1], x \in D,$$

we have $[Z - V]_{\mathcal{C}^{\frac{3}{4},0} L_m([0,1])} \leqslant C(1 + \|b\|^2_{\mathcal{C}^{-1}})(1 + \|\rho\|_{\mathcal{C}^{0,0} L_m([0,1])})$.

Proof. Let $\varphi_t(x) := Z_t(x) - V_t(x)$, $t \in [0,1]$, $x \in D$. For any $(s,t) \in \Delta_{[0,1]}$, $x \in D$, by the projection property of the conditional expectation in $L_m(\Omega)$ (e.g., [BDG25, Lemma 4.2(i)]),

$$\begin{aligned}\|(\varphi_t - \mathsf{E}^s \varphi_t)(x)\|_{L_m(\Omega)} &\leqslant C\|(\varphi_t - P_{t-s}\varphi_s)(x)\|_{L_m(\Omega)} \\ &\leqslant C\Big\|\int_s^t \int_D p_{t-r}(x,y) b(z_r(y))\, dydr\Big\|_{L_m(\Omega)} + (t-s)\|\rho\|_{\mathcal{C}^{0,0} L_m([0,1])}\end{aligned} \tag{25}$$

for $C = C(m) > 0$. To estimate the first term in the right-hand side of the above inequality, we use Corollary 10 with $w \equiv 1$. We get from (14)

$$\Big\|\int_s^t \int_D p_{t-r}(x,y) b(z_r(y))\, dydr\Big\|_{L_m(\Omega)} \leqslant C\|b\|_{\mathcal{C}^{-1}}(t-s)^{\frac{3}{4}}(1 + [\varphi]_{\mathcal{X}}(t-s)^{\frac{1}{2}})$$

for $C = C(m) > 0$. Substituting this into (25) we get for any $(s', t') \in \Delta_{[s,t]}$, $x \in D$

$$\|\varphi_{t'}(x) - \mathsf{E}^{s'}\varphi_{t'}(x)\|_{L_m(\Omega)} \leqslant C(t'-s')^{\frac{3}{4}}\|b\|_{\mathcal{C}^{-1}}\big(1 + \|\rho\|_{\mathcal{C}^{0,0} L_m([0,1])} + [\varphi]_{\mathcal{X}}(t-s)^{\frac{1}{2}}\big).$$

Let $\ell > 0$. Dividing both sides of the above inequality by $(t'-s')^{\frac{3}{4}}$ and taking supremum in the above inequality over all $s', t' \in \Delta_{[s,t]}$, we deduce for any $(s,t) \in \Delta_{[0,1]}$ such that $|t-s| \leqslant \ell$

$$[\varphi]_{\mathcal{X}} \leqslant C\|b\|_{\mathcal{C}^{-1}}(1 + \|\rho\|_{\mathcal{C}^{0,0} L_m([0,1])} + \ell^{\frac{1}{2}}[\varphi]_{\mathcal{X}}) \tag{26}$$

where $C = C(m) > 0$. By assumptions of the lemma the function b is bounded and hence $[\varphi]_{\mathcal{X}} < \infty$. Take now $\ell > 0$ such that $C\|b\|_{\mathcal{C}^{-1}}\ell^{\frac{1}{2}} = \frac{1}{2}$. Then we deduce from (26) that for any $(s,t) \in \Delta_{[0,1]}$ such that $|t-s| \leqslant \ell$ we have $[\varphi]_{\mathcal{X}} \leqslant C\|b\|_{\mathcal{C}^{-1}}(1 + \|\rho\|_{\mathcal{C}^{0,0} L_m([0,1])})$. If $\ell > 1$, then there is nothing more to prove, this inequality implies the lemma's conclusion. Otherwise, if $\ell < 1$, for a general $(s,t) \in \Delta_{[0,1]}$ we choose $N \geqslant 2$ such that $1/N \leqslant \ell < 1/(N-1)$ and define $t_k := s + k(t-s)/N$ for each $k = 0, 1, \ldots, N$. Then $t_{k+1} - t_k \leqslant \ell$ and therefore by the above inequality, triangle inequality, and Jensen's inequality

$$\|\varphi_t(x) - \mathsf{E}^t \varphi_s(x)\|_{L_m(\Omega)} \leqslant CN^{\frac{1}{4}}(t-s)^{\frac{3}{4}}\|b\|_{\mathcal{C}^{-1}}(1 + \|\rho\|_{\mathcal{C}^{0,0} L_m([0,1])}),$$

for $C = C(m, \beta)$. Since $N^{\frac{1}{4}} \leqslant 1 + C\|b\|_{\mathcal{C}^{-1}}$, we get the bound claimed by the lemma. $\square$

Corollary 14. *Let u be a solution to (1) and $\widetilde{v}$ be a solution to (3). Let $m \geqslant 2$, $\mu \in (0, \frac{1}{6})$. Then there exists $C = C(m, \mu) > 0$ such that for any $n \in \mathbb{N}$ we have*

$$\begin{aligned}&[u - V]_{\mathcal{C}^{\frac{3}{4},0} L_m([n,n+1])} \leqslant C(1 + \|h\|^2_{\mathcal{C}^{-1}}); \\ &[\widetilde{v} - V]_{\mathcal{C}^{\frac{3}{4},0} L_m([n,n+1])} \leqslant C(1 + \|h\|^2_{\mathcal{C}^{-1}})(1 + \lambda_n\|u - \widetilde{v}\|_{\mathcal{C}^{0,0} L_m([n,n+1])})\end{aligned}$$

Proof. The bounds follow directly from the definitions of u, $\widetilde{v}$ and Lemma 13. $\square$

Proof of Lemma 1. Bound (5) is a direct combination of Corollary 12 and Corollary 14. Inequality (6) follows immediately from (16) and Corollary 14. $\square$

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